LOCAL BUSINESS SUPPORT RECOVERY

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Today's Agenda

01 The Problem

02 Context & Research Site

03 Methodology

04 Key Data:
Foot Traffic & Visit Motives

05 Key Data:
Spending & Inventory

06 System Analysis Findings

07 Proposed Solution

08 Key Features

09 Implementation Plan

10 Conclusions

The Problem

Small Businesses Are Struggling Post-Pandemic

- 41.18% of mall visitors spend less than before the pandemic
- Foot traffic is unpredictable: Sunday brings 2x more visitors than Saturday
- Small stores operate as isolated agents - no real-time data
- Large chains (Falabella, Éxito) dominate through digital tools; small shops rely on foot traffic alone
- 80% of small stores report fast/very fast inventory rotation → stockout risk is high

Context & Research Site

Why Titan Plaza?

- Located on Av. Boyacá con Calle 80, Bogotá — one of the most visited malls in the city
- 4 floors, 250 stores, daily 8am–10pm
- Attracts middle-income population (strata 3–5)
- Contains a mix of large chains + small independent businesses
- Ideal case study: concentrated, observable, real-world environment

Methodology

How We Analyzed the System

Direct Observation	Surveys	Environmental Correlation
Saturday 9am–12pm & Sunday 12pm–3pm	17 customers + 5 employees	Weather, day of week, promotions
Counted foot traffic at small store areas	Spending habits, visit motives, inventory	Two observation sessions

Key Data: Foot Traffic & Visit Motives

What the Data Tells Us

Day	Enter	Exit
Saturday (9am - 12pm)	343	262
Sunday (12pm - 3pm)	811	309
Total	1154	571

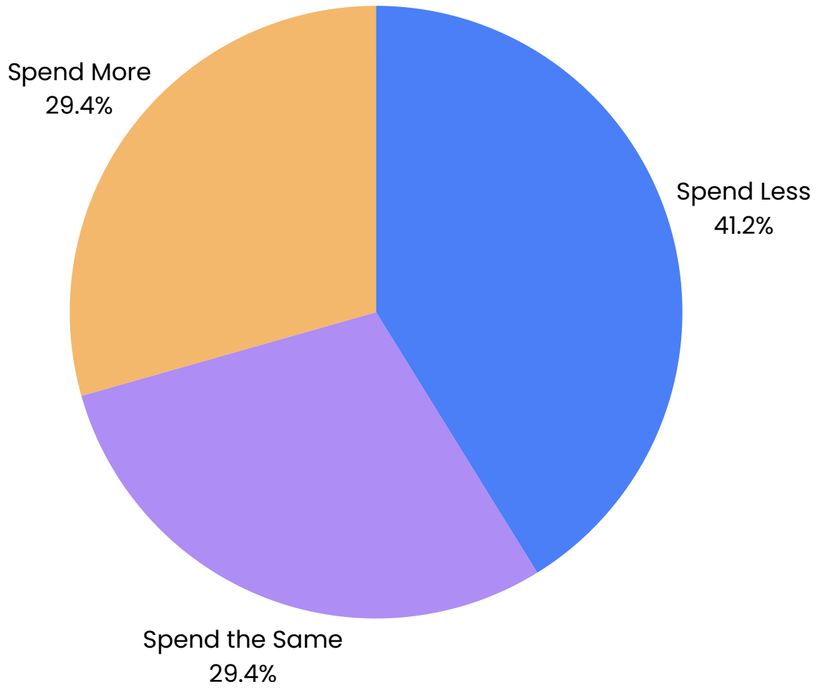
Reasons for visit (top results)

- 29.41% - Other reasons
- 17.65% - Food court / Cinema
- 11.76% - Small / local stores only

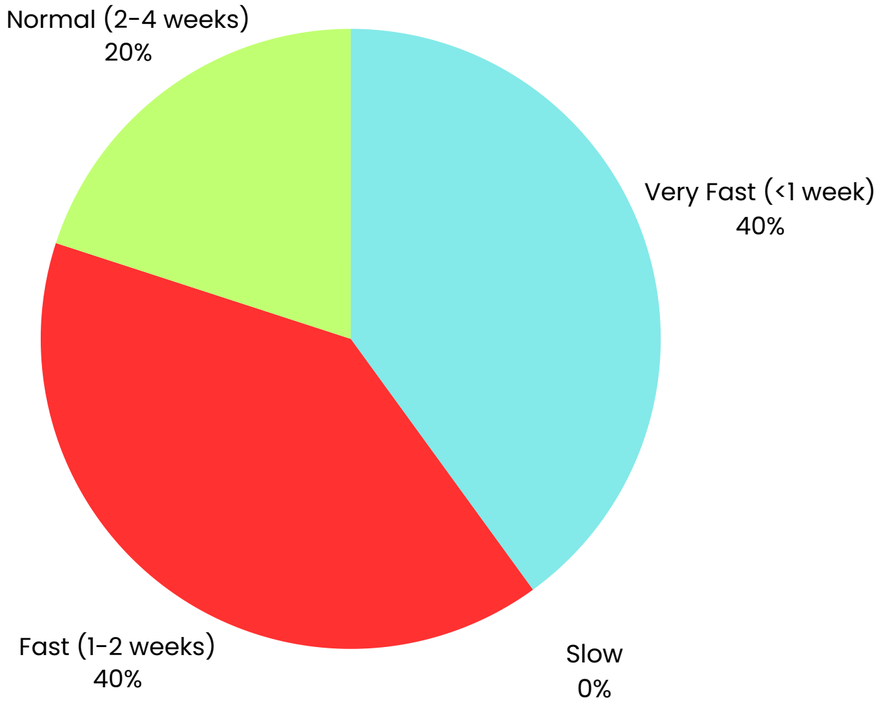
Key Data: Spending & Inventory

The Core Challenge in Numbers

Post-Pandemic Spending:



Inventory Rotation:



System Analysis Findings

What's Really Broken?

Visibility Gap

Small stores have no digital presence customers go straight to large chains or e-commerce.

Feedback Loop Problem

One bad experience (item out of stock) permanently redirects customers to anchor stores.

Informational Disconnect

Each shop operates in isolation. No shared data on traffic patterns, demand trends, or customer needs.

Proposed Solution

Local Business Support System

A modular analytics platform that integrates customer, inventory, and environmental data to help small businesses make real-time, data-driven decisions.

IMPUTS

- Weather
- Customer behavior
- Economic conditions
- Business data

PROCESSING

- Inventory management
- Data analysis
- Recommendation engine
- Reporting module

OUTPUTS

- Customer satisfaction
- Revenue reports
- Operational insights

Key Features

What the Platform Does

- **Owner Dashboard:** Real-time metrics - sales, traffic, inventory turnover
- **Inventory Alerts:** Notifies owners before stock runs out
- **Automated Promotions:** Triggered by traffic peaks and low-sales periods
- **Traffic Prediction:** Based on day of week, weather, and historical patterns
- **Product Recommendations:** Personalized suggestions based on purchase trends

Implementation Plan

How We'll Build It

Phase 1

DB schema + core logic + basic dashboard (read-only)

Phase 2

Sales module + inventory management + alert system

Phase 3

Demand forecasting + recommendation engine

Phase 4

Pilot with selected stores → feedback → full rollout + training

Conclusions

The platform directly addresses the informational disconnect that prevents small businesses from competing. Personalized promotions tackle the 41.18% spending drop; inventory alerts prevent stockouts where 80% of products move in under 2 weeks; and the dashboard gives owners the same data advantage that large chains already have.

Final Bullets

- Data-driven, not intuition-driven
- Scalable to other malls
- Designed for real constraints (connectivity, tech literacy, legal compliance)

Future Work

- Loyalty programs & customer-facing features
- Cross-mall benchmarking
- Improved ML forecasting as data accumulates

Thank You

Better design, better living.